Question 1

A ferromagnetic core with a depth of the of 5 cm and relative permeability of 1550 is shown in Figure 1.1. The air gap on the right side of the core is 0.67 cm with effective area 8 % larger than their physical size due to fringing effects. Given current i = 15 A.

a. Find the total reluctance of the core and air gap. (9 marks)
b. Find the total flux produced by this core. (2 marks)
c. Find the flux density of the left side of the core. (4 marks)

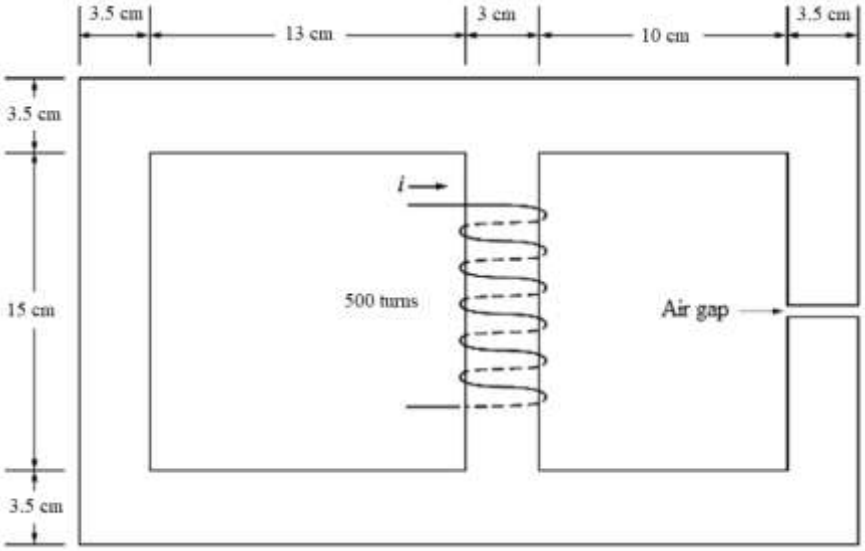


Figure 1

Question 2

A shunt DC generator generates a rated output of 2.3 kW and 120 V to the load while it is rotating at 1200 rpm. Given that the fixed rotational losses of the generator is 620 W, the total voltage drop across the brushes is 2.8 V, and the resistances of shunt field and armature winding are 32 Ω and 0.5 Ω, respectively. At the rated condition, find:

a. The shunt field current, I_{sh}. (2 marks)
b. The armature current, I_a. (4 marks)
c. The generated emf, E_G. (3 marks)
d. The mechanical input power, P_{in} if all the losses are taken into account. (6 marks)
e. The shaft torque. (3 marks)
f. The efficiency of the generator. (2 mark)

Question 3

A series DC motor of 16-hp and 250-V has an armature resistance of 0.5 Ω and a series field resistance of 0.11 Ω. The input current is 62 A during full load with rated speed of 950 rpm. The core and the friction/windage losses are 400 W and 420 W, respectively. The total voltage drop across the brushes is 2V. The E_a vs I_f plot of this motor at speed of 1100 rpm is shown in Figure 2.

1. Calculate the copper, brush and stray losses of the motor at full load. (8 marks)
2. Determine the efficiency of the motor at full load. (2 marks)
3. If the armature current is increased to 90 A, find the speed and efficiency of the motor. Assume that the rotational losses are constant. (10 marks)

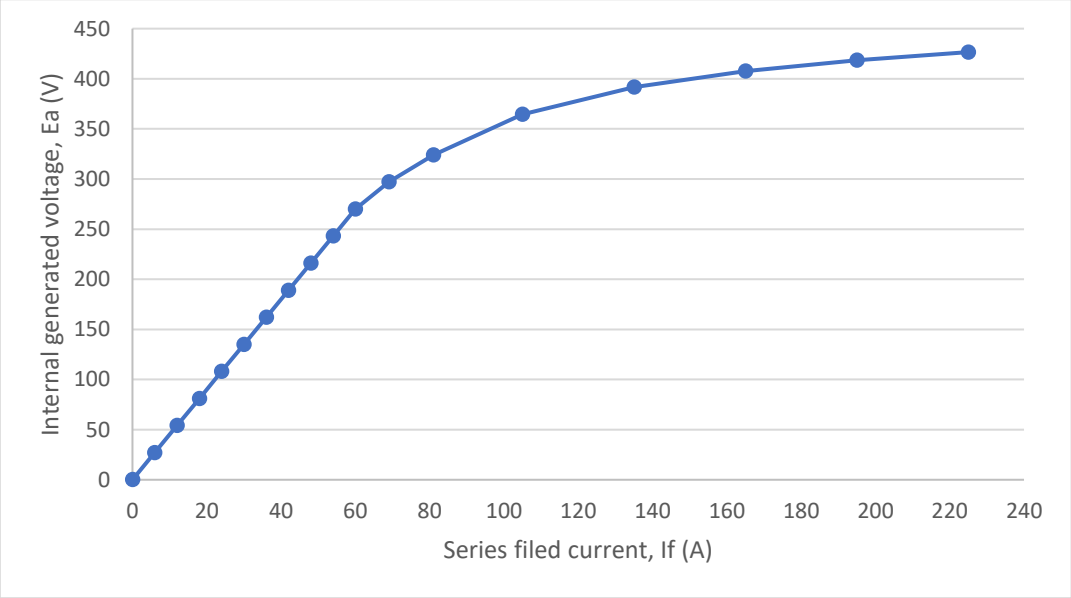


Figure 2

Question 4

a. The equivalent circuit parameters of a 460 V_L, 8-pole, and 60-Hz Y-connected induction motor has the following impedances:

   $R_1 = 0.77\ \Omega$ $X_1 = 1.3\ \Omega$ $X_M = 19\ \Omega$

   $R_2 = 0.8\ \Omega$ $X_2 = 0.72\ \Omega$

   At slip = 2.6%, the friction and windage losses, $P_{f,w} = 80$ W, stray losses $P_{stray} = 20$ W, and core losses $P_{core} = 2.25$ kW, calculate

   i. The synchronous and motor speeds. (2 mark)

   ii. The stator current I_1, considering the effects of magnetizing branch X_m, where $I_1 \neq I_2$ (4 marks)

   iii. The power factor, (lagging, leading or unity) (2 marks)

   iv. The electromechanical P_{mech} and output power P_{out} (6 marks)

   v. The efficiency of the motor. (2 marks)

b. An induction motor has a short circuit current that is 6 times the full load current. At full load, the slip is 4.3%.

   i. The prevent high current from damaging the motor, determine the suitable transformation ratio on an auto-transformer starter to start the motor with 45% full load torque. (3 marks)

   ii. Determine the current drawn from the supply during starting with this auto transformer, if the full load current is 150 A. [CLO3-PLO2:C4] (1 mark)

Question 5

A 1000 kVA, 2.3 kV, 3-phase 8 pole Y-connected generator is connected to an infinite bus with line voltage of 3 kV. Given the internal induced voltage and synchronous reactance of the generator is 2 kV and 11 Ω per phase, respectively.

a. Determine the mechanical angle if the generator is delivering active power of 600 kW to the terminals. (5 marks)

b. Calculate the phase voltage, internal induced voltage, voltage across synchronous reactance, armature current, and the angles between phasors, then illustrate the phasor diagram and label all relevant values. (10 marks)

Question 6

An 8-pole, 470-hp, Δ-connected synchronous motor is powered by a 480-V_L, 50 Hz AC supply. It has a synchronous reactance of 0.85 Ω and it operates at 0.92 PF lagging. The friction, windage, and core losses amount to 7% of rated power. Assume electrical losses are negligible.

a. Calculate I_A, E_b, δ and θ at its rated conditions and illustrate the phasor diagram. (10 marks)
b. Determine the induced torque and the maximum possible induced torque of the motor for this field current setting. (5 marks)

Question 7

The effective resistance and reactance of the main windings of a capacitor-start induction motor are 2.1 Ω and 7.4 Ω, respectively. The auxiliary windings of same motor have effective resistance and reactance of 5.3 Ω and 1.6 Ω, respectively. These values are measured under locked-rotor conditions. If the terminal voltage, V_L is 110 V,

a. Illustrate the equivalent circuit and determine the magnitude of real, reactive and apparent powers from the main and auxiliary windings. (10 marks)
b. Calculate the phase angle between I_a and I_s. (4 marks)
c. Calculate the line current I_L. (4 marks)
d. Determine the power factor under locked-rotor conditions. (2 marks)

Question 8

A 100 V, 60 Hz, four-pole, single-phase induction motor has the following impedances:

$R_1 = 1.50\ \Omega$; $X_1 = 2\ \Omega$; $X_M = 45\ \Omega$

$R_2 = 2.20\ \Omega$; $X_2 = 2\ \Omega$

The motor's rotational losses are 60 W and may be assumed constant over the normal operating range of the motor. Calculate the P_{in}, P_{AG}, P_{conv}, and P_{out} of this motor if the slip is 0.06. [CLO3-PLO2:C4] (15 marks)